Exercises: Loading and Inspecting the `nlsy97_emp_hed.parquet` Dataset

In the following, let us use the NLSY97 Monthly Employment and College Enrollment Status data file (available on Moodle): nlsy97_emp_hed.parquet.

I. Loading Data

(a) Let us begin with loading the necessary packages:

```
# external (third-party) packages
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# packages of the standard library
from pathlib import Path
```

(b) Next, let us set the project directory. I name it `ROOT` (root directory), i.e. the folder containing the subfolders and files needed for the project.

We can then define the project's root directory as a string or as an explicit `Path` object (from the `pathlib` module):

```
# Your path will almost certainly differ from the one below.
ROOT = Path(r'C:\Users\PC_DS_ECON_5\Desktop\data-analytics-python-wise2627')
ROOT
```

(The `r` before the string indicates a *raw string*. In ordinary Python strings, certain character sequences have special meanings. For example, `'\n'` represents a newline character. Raw strings are particularly useful for writing Windows paths, which contain backslashes.)

You should set `ROOT` according to the location of the project folder on your own computer.

If we are not sure in which folder we currently are, we can run the following code to find it out:

```
cur_path = Path.cwd()
print(cur_path)
```

(c) Let us specify the path of the folder in which we store our data file:

```
DATA = ROOT / 'local' / 'data'
```

It is up to you how you name the folder and whether you put it in a folder called `local`.

(d) Let us read in the `nlsy97_emp_hed.parquet` file using `pandas.read_parquet()`:

```
df = pd.read_parquet(DATA / 'nlsy97_emp_hed.parquet')
```

(To read the file, one needs a Parquet engine, such as the one provided by the `pyarrow` package. The package can be installed in the usual way with `pip install pyarrow` after activating the virtual environment.)

II. Inspecting the dataset

(a) Inspect the object by running the following commands one after the other:

```
# type of the object
print('type of object: ', type(df))
# column names
print('variables: ', df.columns)
# column dimensions
print('dimensions: ', df.shape)
```

Question: What type of object is `df`? How many columns/variables are there? How many rows/observations are there?

Run the following code to display the first ten rows/observations of the dataset:

```
df.head(10)
```

(b) Run the following code to display the number of observations for each year:

```
df.value_counts(subset=['year'], normalize=False)
```

Which year has the most observations?

Run the following code to calculate the relative frequencies of employment status categories in June 2009:

```
(
    df
        .loc[df['year_month'].eq(pd.Period('2009-06', freq='M')),:]
        .value_counts(subset=['emp_category'], normalize=True)
)
```

What share of observations with a known employment status were classified as unemployed?

(Bonus question: Does this share correspond to the official unemployment rate? Why or why not?)

(c) What is the unit of observation? Put differently, which of the following combinations of variables uniquely identifies each row/observation?

- `person_id`
- `person_id` X `year`
- `person_id` X `year_month`

How would you use the `value_counts()` method to find it out?